UNIT 8 · RAPID REVISION

Formula and Fact Sheet

Materials

Item	Value
Mild steel	< 0.3% C — ductile, weldable (frames, panels)
Medium-carbon steel	0.3–0.6% C — shafts, axles, gears
High-carbon steel	0.6–1.4% C — springs, shares, cutting edges
Cast iron	2–4% C; grey, white, malleable, SG (Mg-treated)
Stainless steel	≥ ~11% Cr
Boron in tillage steel	~0.0005–0.003% — raises hardenability
Share/point hardness target	50–60 HRC
AR400 plate	BHN 360–440 (AR200 170–250 … AR600 560–625)
Aluminium density	~2.7 g/cm³ (⅓ of steel)
Brass / bronze	Cu–Zn / Cu–Sn
Hardness tests	Brinell (ball), Rockwell C (diamond cone), Vickers (diamond pyramid)
Pig iron / steel / cast iron furnaces	Blast furnace / BOF or EAF / cupola

Workshop, press and machining

Item	Value
Tap / die / reamer	Internal threads / external threads / finish a hole
Blanking vs piercing	Blanking: punched piece is the product; piercing: the hole is wanted
Blanking force	$F = L \times t \times \tau_s$
Punch–die clearance	~5–10% of thickness per side
Compound die	Two operations (blank + pierce) in one stroke
Taylor's equation	$V \cdot T^n = C$
Taylor exponent n (source)	HSS 0.16, carbide 0.30, ceramic 0.40
Jig vs fixture	Jig guides the tool (drilling); fixture only holds (milling)
Locating principle	3-2-1 (six-point)
Plug / snap gauge	Holes / shafts

Casting and welding

Item	Value
Cope / drag	Top / bottom half of the flask
Riser	Feeds shrinkage
Draft allowance	Easy withdrawal of the pattern
Die casting	Permanent steel die, high pressure, non-ferrous, mass production
Investment casting	Lost-wax
Oxy-acetylene flame	~3200 °C
Brazing vs soldering	Filler above vs below 450 °C
TIG electrode	Non-consumable tungsten
HAZ	Unmelted but altered; ~5–15 mm; often weakest
NDT map	PT surface (any metal), MT ferromagnetic surface, RT internal (film), UT internal (sound)

Materials (Topic 1)

- Ferrous: mild < 0.3% C (ductile, weldable), medium 0.3–0.6% C (shafts/gears), high 0.6–1.4% C (springs, shares); ↑C → ↑hardness, ↓ductility/weldability.
- Cast iron C 2–4%: grey (machinable/damping), white (hard/brittle), malleable & SG/ductile (tough); melted in a cupola. Steel via blast furnace → BOF/EAF.
- Alloying: Cr (hardness/corrosion, stainless ≥ 11%), Ni (toughness), Mn (Hadfield, impact wear), B (hardenability — tillage steel), Mo/V (high-T strength, grain refinement).
- Non-ferrous: Al (light ~2.7 g/cm³, radiators/pistons), Cu (conductor), brass Cu–Zn (fittings), bronze Cu–Sn (bearings/bushes), Zn (galvanising).
- Composites: matrix + reinforcement; FRP (GFRP/CFRP) tanks/guards/hoods; MMC/hard-facing (WC) for wear edges; high strength-to-weight, corrosion-free.
- Shares/tines → boron/high-C steel 50–60 HRC + hard-facing; hardness tests BHN/HRC/HV; AR grades BHN 170–625; heat treatment delivers the properties.

Workshop, press, machining, jigs and gauges (Topics 2–4)

- Prototype flow: design → marking → cutting → forming → joining → machining → finish/assemble; fitting tools = vice, files, hacksaw, taps/dies, reamers.
- Press: shearing (straight cut), BLANKING (punched piece = product), PIERCING (hole wanted, slug = scrap); $F = L \cdot t \cdot \tau_s$; clearance 5–10% of t per side; simple/compound/progressive dies.
- Machining: rake angle (chip flow/force), clearance angle (no rubbing); turning/drilling/milling/grinding; speed–feed–depth; tools HSS < carbide < ceramic < CBN/diamond.
- Taylor: $V \cdot T^n = C$; n ≈ HSS 0.16, carbide 0.30, ceramic 0.40; higher n = life less speed-sensitive; wear = flank/crater.
- JIG holds + GUIDES the tool (drill bushes); FIXTURE holds only (milling/turning); 3-2-1 locating. GO/NO-GO limit gauges; plug (holes), snap/ring (shafts); Taylor's principle — GO = all dimensions at max-metal, NO-GO = one dimension at min-metal.

Casting, welding and workshop planning (Topics 5–7)

- Sand casting: cope + drag, pattern, sprue/riser/core, parting line; allowances = shrinkage, machining, draft, distortion, shake; works for ferrous too.
- Die casting: permanent steel die, HIGH pressure, non-ferrous (Al/Zn/Mg), hot- vs cold-chamber, mass production; investment = lost wax.
- Welding: fusion (arc/gas/resistance) vs pressure/solid-state; SMAW/MIG/TIG/SAW; brazing > 450 °C, soldering < 450 °C; oxy-acetylene ~3200 °C.
- Zones: weld metal → HAZ (5–15 mm, weakest/crack-prone) → base metal; defects porosity/slag/lack of fusion/undercut/cracks (hot vs cold); residual stress → distortion.
- NDT: VT, PT (surface, any metal), MT (ferrous surface/near-surface), RT (internal, film), UT (internal, sound). Destructive = tensile/bend/impact/hardness/macro.
- Workshop: lathe/drill/mill/shaper/grinder/press/shear/welding sets; plan for straight material flow, safety, lighting; separate hot/noisy work from precision work.

Further reading

- Callister — Materials Science and Engineering; Rajput — Engineering Materials.
- Hajra Choudhury — Elements of Workshop Technology, Vols I & II.
- Kalpakjian & Schmid — Manufacturing Engineering and Technology.
- AWS Welding Handbook (processes, weld metallurgy, inspection).
- BIS/IS material specifications for tillage-tool shares, tines and discs.
